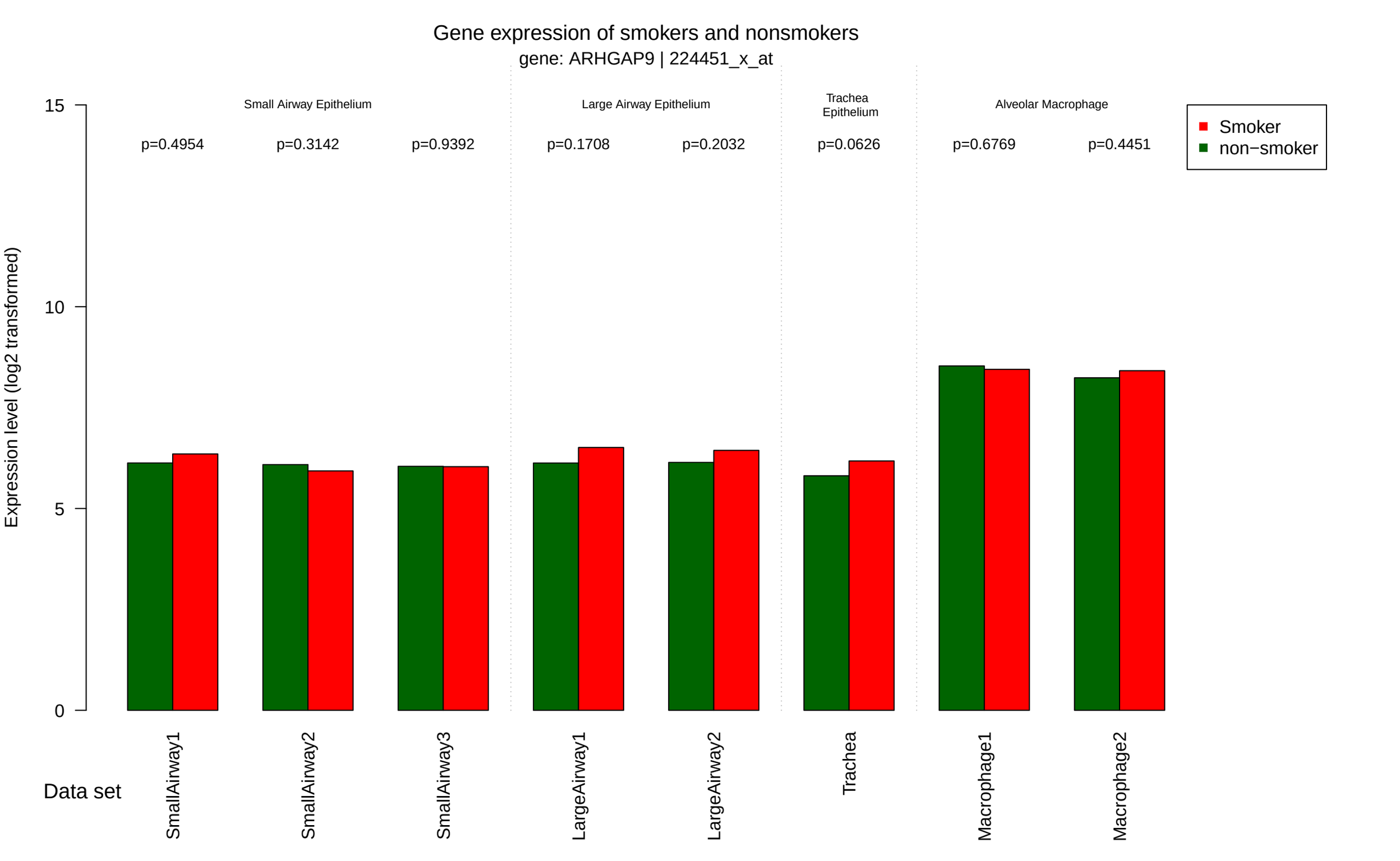
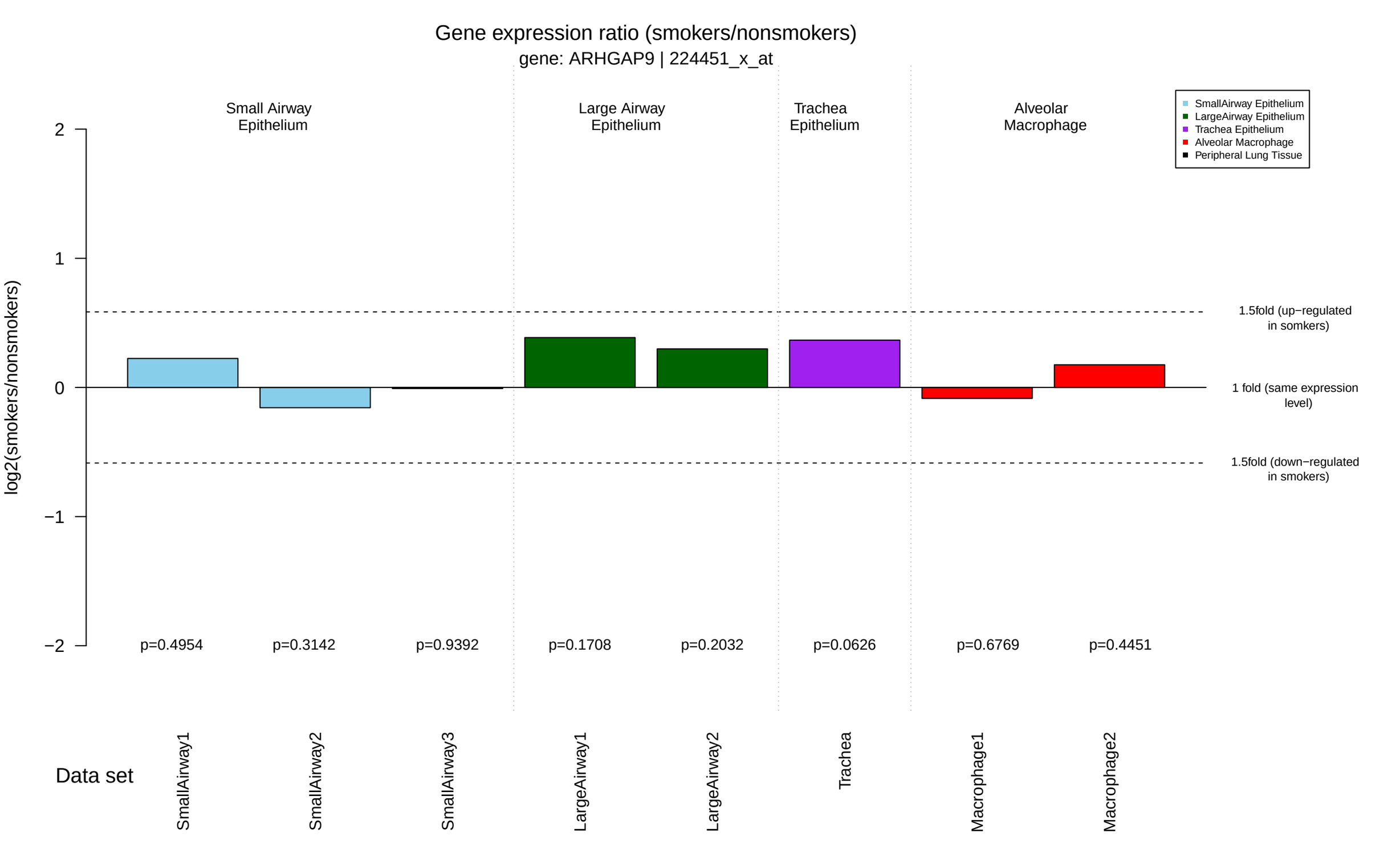
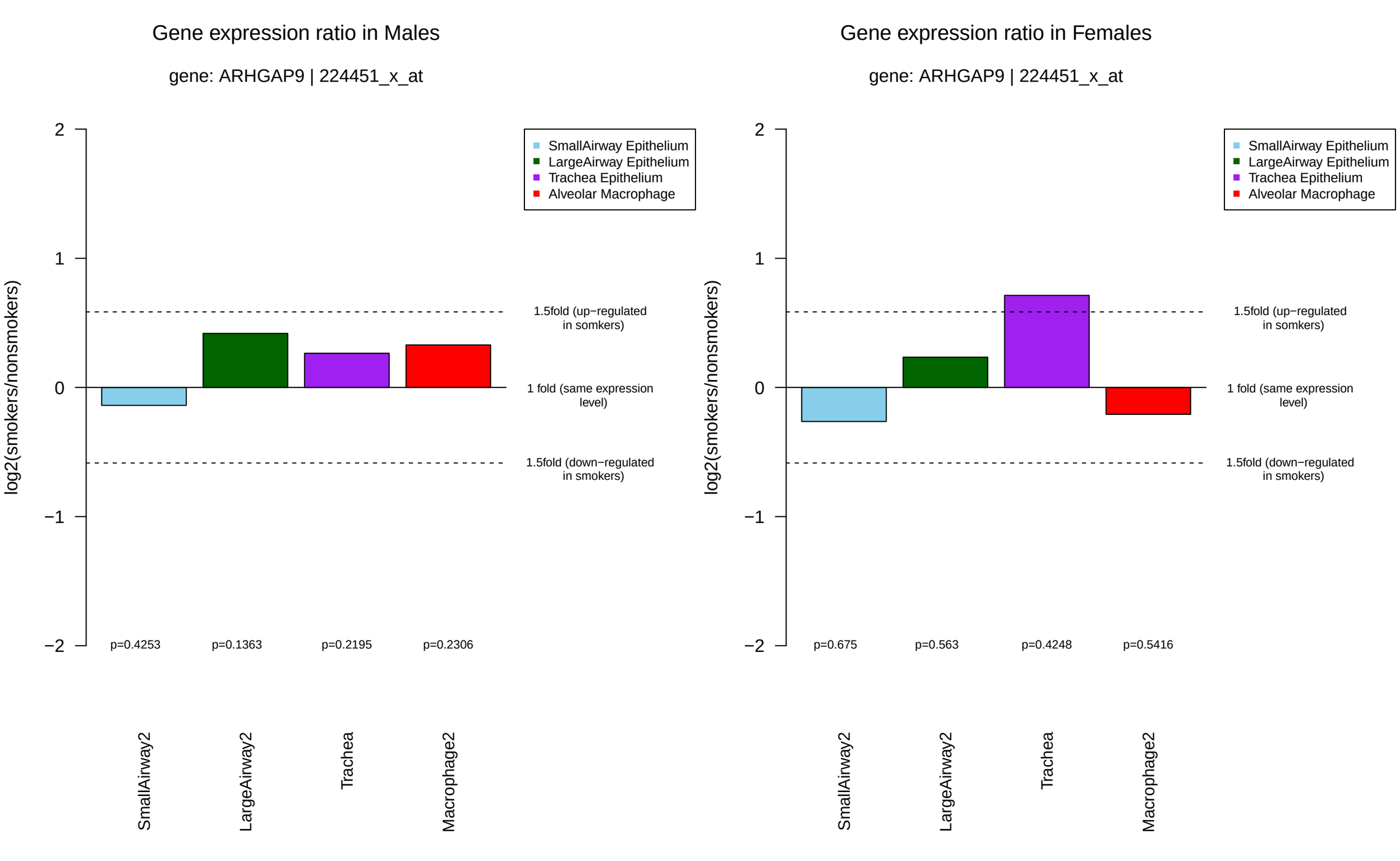
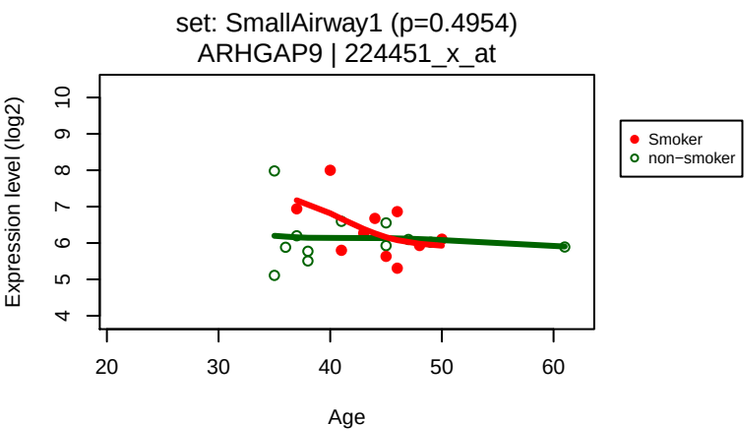
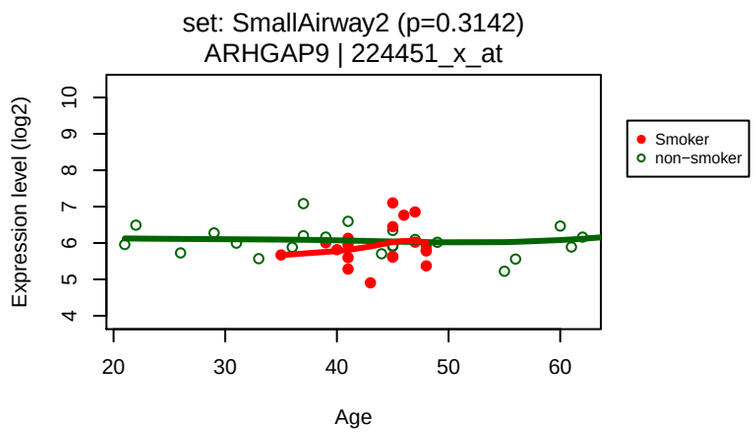
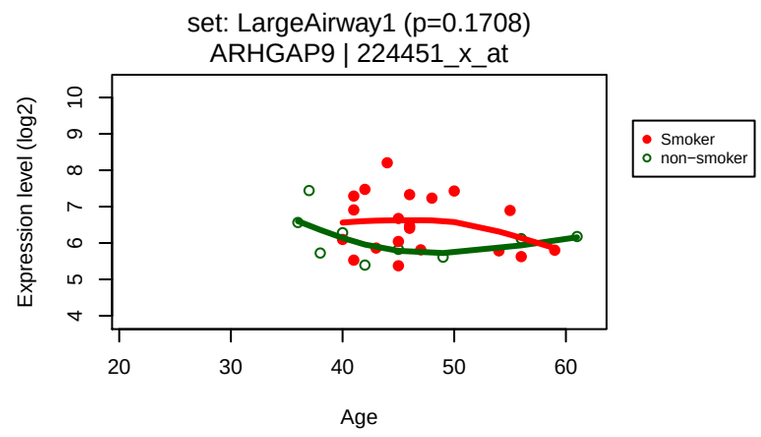

Gene expression of smokers and nonsmokers
gene: ARHGAP9 | 224451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4954
p=0.3142
p=0.9392
p=0.1708
p=0.2032
p=0.0626
p=0.6769
p=0.4451
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

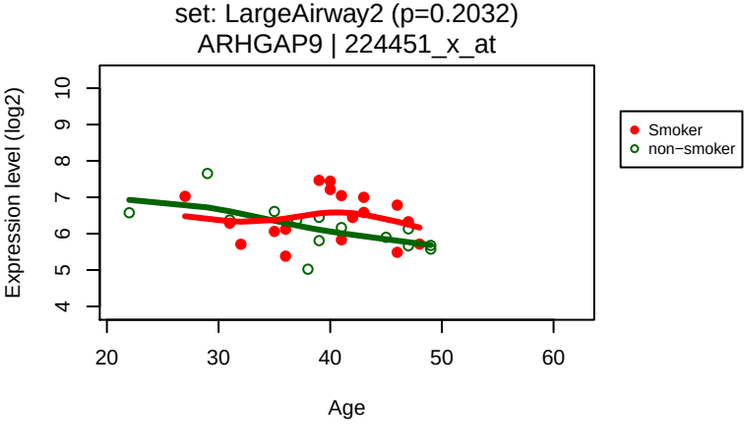

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP9 | 224451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4954
p=0.3142
p=0.9392
p=0.1708
p=0.2032
p=0.0626
p=0.6769
p=0.4451
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

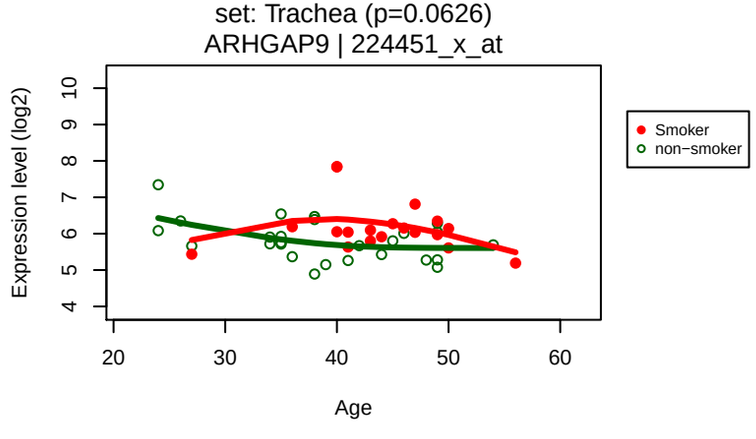

Gene expression ratio in Males
gene: ARHGAP9 | 224451_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4253
p=0.1363
p=0.2195
p=0.2306
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP9 | 224451_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.675
p=0.563
p=0.4248
p=0.5416
SmallAirway2
LargeAirway2
Trachea
Macrophage2

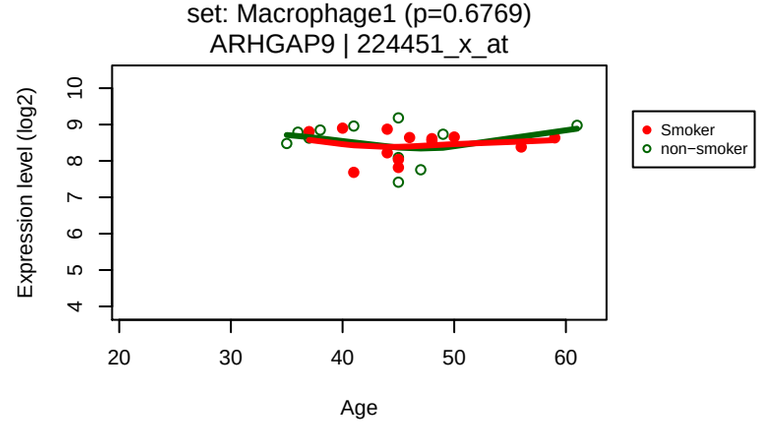

set: SmallAirway1 (p=0.4954)
ARHGAP9 | 224451_x_at
Expression level (log2)
Age
Smoker
non−smoker

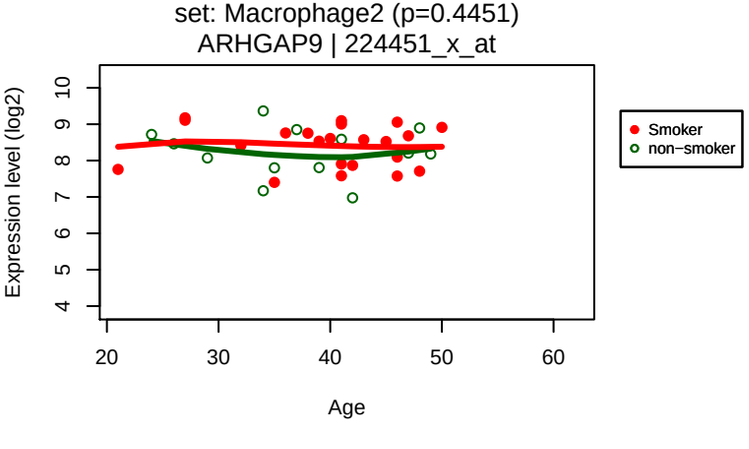

set: SmallAirway2 (p=0.3142)
ARHGAP9 | 224451_x_at
Expression level (log2)
Age
Smoker
non−smoker

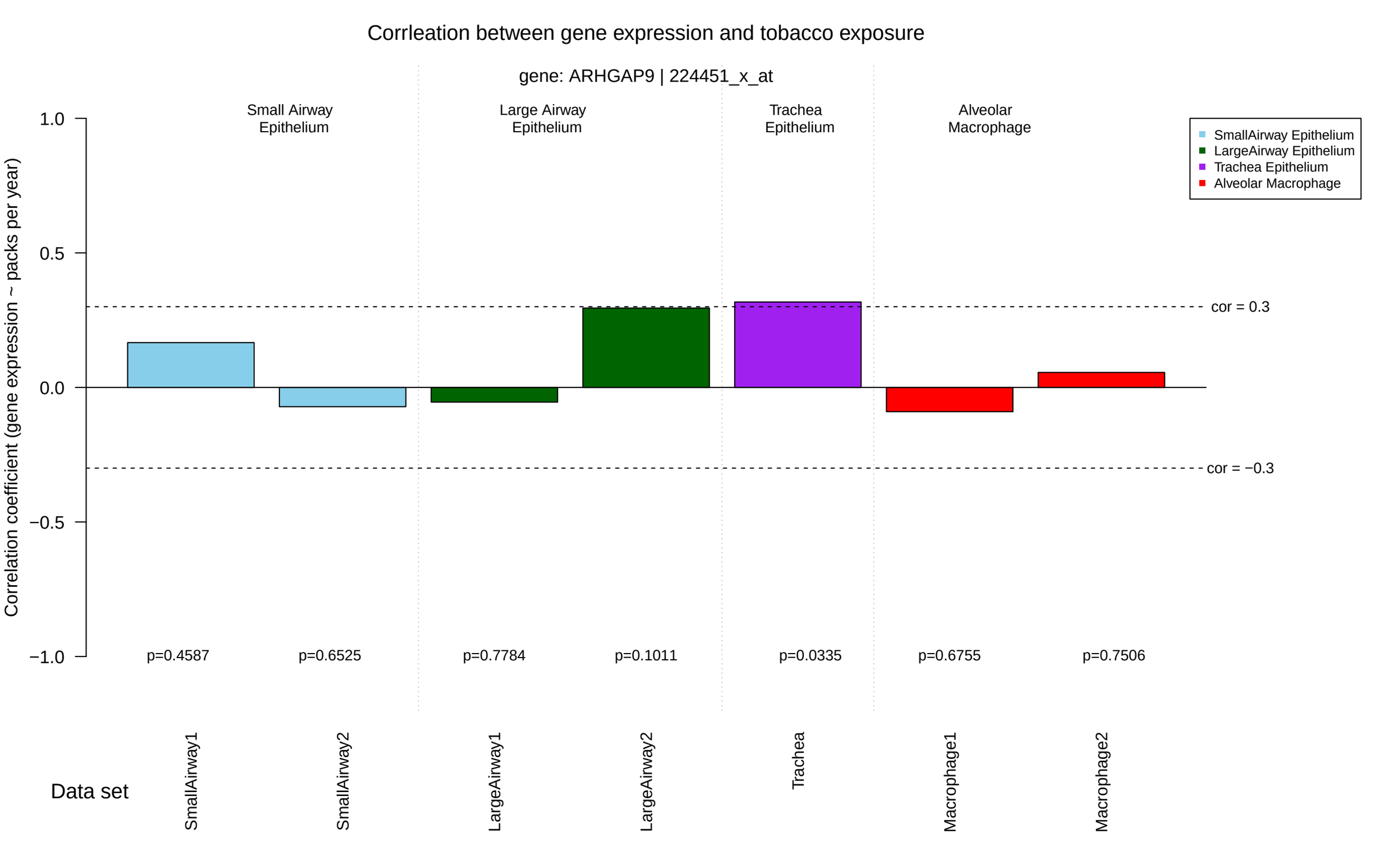

set: LargeAirway1 (p=0.1708)
ARHGAP9 | 224451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2032)
ARHGAP9 | 224451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0626)
ARHGAP9 | 224451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6769)
ARHGAP9 | 224451_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4451)
ARHGAP9 | 224451_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP9 | 224451_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4587
p=0.6525
p=0.7784
p=0.1011
p=0.0335
p=0.6755
p=0.7506
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2